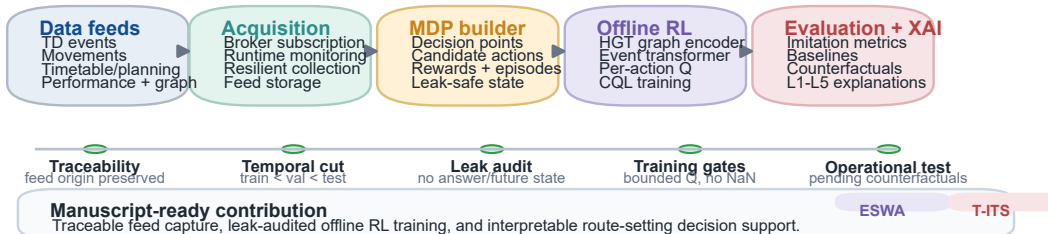


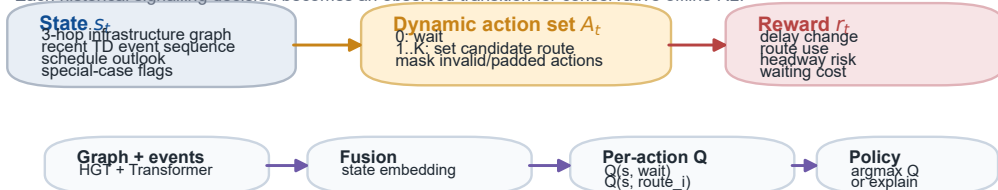
a End-to-end RailRL system

Public operational feeds are transformed into leak-audited offline RL transitions and explainable route-setting recommendations.



b Decision-point mechanism

Each historical signalling decision becomes an observed transition for conservative offline RL.



Observed transition: $(s_t, a_t, r_t, s_{t+1}, done_t)$ from canonical episode order

Training uses observed next states only; counterfactual rollouts are reserved for evaluation and explanation.

Figure 1. End-to-end RailRL architecture for route-setting decision support.